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| **Fiscal Year:** | FY 2025 | **Task Last Updated:** | FY 09/25/2025 |
| **PI Name:** | Konopka, Uwe Ph.D. | | |
| **Project Title:** | Complex Plasma under Microgravity, Utilizing the International Space Station Experiment “PK-4” and Beyond | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Fundamental physics | | |
| **Joint Agency Name:** | National Science Foundation | **TechPort:** | No |
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| **Human Research Program Risks:** | None | | |
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| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 36849-0001 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS |
| **Start Date:** | 06/15/2022 | **End Date:** | 09/30/2025 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 2 |
| **No. of PhD Candidates:** | 4 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 3 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | Callas, John | **Contact Phone:** | |
| **Contact Email:** | john.l.callas@jpl.nasa.gov | | |
| **Flight Program:** | ISS | | |
| **Flight Assignment:** | ISS<br>NOTE: End date changed to 09/30/2025 per I. Hahn/JPL (Ed., 6/18/25).<br>NOTE: End date changed to 06/15/2025 per J. Callas/JPL (Ed., 8/29/24).<br>NOTE: New contract number is JPL #1679198 per J. Callas/JPL. No change in scope (Ed., 8/20/24).<br>NOTE: End date changed to 12/31/2023 per U. Israelsson/JPL (Ed., 1/6/22)<br>NOTE: End date changed to 12/31/2021 per U. Israelsson/JPL (Ed., 4/13/21) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Thomas, Edward Ph.D. ( Auburn University )<br>Williams, Jeremiah Ph.D. ( Wittenberg University ) | | |
| **Grant/Contract No.:** | JPL 1679198 ; JPL 1571699 | | |

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| **Performance Goal No.:** | |
| **Performance Goal Text:** | |
| **Task Description:** | Complex plasmas are four-component systems that contain electrons, ions, neutral atoms, and charged microparticles (i.e., the dust particles). Depending upon the experimental conditions, the microparticles in these systems can show solid-like, liquid-like, or gas-like properties. However, ground-based experiments are subject to gravitational forces that often dominate the behavior of the systems under investigation leading to sedimentation and anisotropic effects. Therefore, in order to study large-scale complex plasmas and the influences of the weaker forces that are masked by the gravitational force, experiments have to be performed under “microgravity” conditions. Over the past decade, several complex plasma experiments have been performed using the International Space Station (ISS). The latest of these research instruments, “Plasma Kristall 4” (PK-4), is the first microgravity laboratory that is optimized to study fluid phenomena and fundamental effects such as particle charging, screening, and the influence of a directed ion flow. Additionally, since PK-4 is a multi-user facility, we have a unique opportunity to use this instrument to study ion/particle and particle/particle interactions that we otherwise would not be able to address in ground-based experiments.<br>The intellectual merit of this research project is centered on three major research topics.<br><br>Topic 1: Charging and interaction of microparticles in the plasma. The most important measurement in a complex plasma is the measurement of the charge on the particles. However, direct experimental measurement of the charge is difficult, because it usually relies on additional assumptions about experimental parameters such as the plasma electric field and on the validity of an assumed particle interaction type. The microgravity environment of PK-4, where the dust is suspended in the bulk plasma, allows experiments that can potentially lead to a higher accuracy in the particle charge determination.<br><br>Topic 2: Ion-dust interactions and the thermal properties of dusty plasmas. A key observation in complex plasmas is the unusually high dust kinetic temperature. It has been postulated that ion-dust interactions are largely responsible for heating the dust particles. The microgravity environment of the PK-4 laboratory, in combination with the tools to actively manipulate the ions in the plasma, provides a unique platform for studying the transfer of energy between ions and the microparticles in the plasma.<br><br>Topic 3: Particle transport and heating in waves. The low charge to mass ratio of the dust leads to characteristic frequencies for the dust component that are on the order of a few Hz. This, coupled with the finite size of the dust grains, allows researchers a unique opportunity to directly image plasma physics phenomena and particle transport at the kinetic level. In the microgravity environment of PK-4, it will be possible to measure the transport properties of particles during their initial injection into the bulk plasma, follow the particles as they form collective structures (e.g., waves), and study how ion flow modifies the thermal properties of these waves.<br><br>Topic 4: How does an initially liquid system transition under compression to a crystalline system? As the inter-particle spacing of the individual particles of a particle cloud is reduced during a compression, the coupling (ratio of interaction energy to thermal energy) between the particles in the cloud will increase. As a result it is expected that the system will change its internal structure from liquid-like (disordered) to crystalline (ordered). Our goal is to find the critical characteristics (temperature, density) of the system, when this transition happens.<br><br>Topic 5: Is the system stable when it is compressed? If not, what causes the instability/waves? Where is the energy input originated?<br><br>This proposed research project has several opportunities to have a broader impact: (1) through on- and off-campus outreach activities and the training of undergraduate and graduate students, this project will cultivate the next generation of plasma scientists; (2) through the development of new ground-based laboratory instruments (e.g., plasma sources, diagnostics) that will complement the proposed research and contribute to the development of next-generation dusty plasma microgravity experiments; and (3) through the development of new, advanced software tools for image analysis and particle tracking that will support the proposed studies and be offered as community-wide research tools. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The research performed under this project falls in the main category of fundamental research. Often, findings obtained by performing fundamental research show their true impact for the society only after many years. However, the study of dusty/complex plasma can benefit the overall research infrastructure in many ways already in the timely vicinity of any discovery made, mainly as a result of the strong interdisciplinary nature of this research. Studying complex plasma topics such as wave phenomena, heating, the spatial separation of phases, their interaction or the transition between phases such as from fluids to crystals all being analyzed down to the scale of individual particles can lead to fundamental finding that might be immediately applied to many other research fields such as solid state physics, fluid physics, plasma physics, astrophysics. As the specific research performed within the scope of this project make use of the microgravity (free-fall) environment supported by the national laboratory in space, the International Space Station (ISS), further utilizing an experiment that is built by ESA, highly international cooperation activities are a direct result. It is likely that the effort to work on the same fundamental questions in a cooperative acting international team will ultimately benefit each involved nation also beyond just the research. Some aspects of the specific research topics within the scope of this project involves the development of universal analysis software, hardware or analysis tools that can be used under different research situations too and some of the more specific technical tools that are developed such as a plasma impedance probe might even help to overall improve technology that can be used for industrial purposes. Understanding the behavior of dust in a plasma environment, especially in a low gravity environment, is further highly advantages for example in mitigating the effect and connected hazards of dust for future, manned space mission to the Moon and/or Mars. Our activities on the described project indeed triggered follow up activities towards lunar dust mitigation research and related technology developments. As part of the general research mission, we also train the next generation of engineers and scientist who can in the course of their future career produce a large benefit for the overall economic situation of the nation and beyond. |

The following report covers the last three years of our project towards the conclusion of the project in September 2025. The project relates to the research agreement RSA-JPL-1571699 and its extension RSA-JPL-1679198. The team working on the project has been built upon researchers and students of Auburn University and Wittenberg University. While a major part of the work of the Principal Investigator (PI) and Co-Investigators (CoIs) has been to guide the research and development activities and supervise students, a major load of the scientific activities have been carried out by graduate and undergraduate students over the whole project. The former Auburn graduate students, Dylan Funk and Brandon Doyle, who were major contributors to the scientific activities over a wide part of this project, concluded their work in the form of PhD thesis during the reporting period. We added the graduate student Mason Sake to the project team. The CoIs have partly reduced some of their activities more recently to address their new, additional administrative duties as dean (E. Thomas), or program manager for the National Science Foundation/NSF (J. Williams), while the PI has increased his engagement to balance the workload. During the reporting period,we extended our cooperation with domestic and international institutions, such as:
• The leading PK-4 core group of the International Space Station (ISS) experiment “Plasma Kristall 4” (PK-4), initially localized at the German Aerospace Center (DLR) in Wesslingen, Germany, now mainly operating from Giessen University. This group operates the ISS based experiment PK-4 as well as a ground based reference module. It also performs related parabolic flight experiments with a PK-4 reference setup and potential future successor complex plasma experiments; • A team at Baylor University in Texas. Here a PK-4 equivalent ground setup is accessible to our group within the cooperation. In addition, we usually proposed PK-4 experiments together to make use of synergies between both groups; • Member of the University of Greifswald, Hochschule of Mittweida and the German Aerospace Center (DLR), Germany. The team at University of Greifswald hosts the core development team for the PK-4 successor experiment, COMPACT. Hochschule Mittweida is performing developments for a core component of COMPACT, the radio frequency plasma generation system. The German Aerospace Center was the employer of the COMPACT and PK-4 core teams before the latter were partly split up to Greifswald and Giessen. The DLR agency is the lead organization for building COMPACT. [Ed. Note: COMPACT is a multi-particle experiment facility for the ISS (the COMplex PlAsma faCiliTy). Uwe Konopka has been participating in and chairing the COMPACT science definition team (SDT), a board initiated to 1) collect and organize experiment ideas by the international complex plasma community for COMPACT; to 2) compile the experiment requirements based on the best experiment ideas; and to 3) support the COMPACT development and breadboard activities in general. As part of our COMPACT development and organization activities, we built new and/or intensified collaborations with the German Universities of Mittweida, Giessen, and Greifswald. In particular, the PI co-supervised the hardware development activities for the COMPACT plasma generation system, and participate in the development of the manipulation and diagnostic systems and the COMPACT facility in general. The SDT also has led to regular bi-weekly information exchange with various agencies and collaboration with COMPACT relevant industrial partners. With the extension (RSA-JPL 1679198) to the initial project (RSA-JPL 1571699), the scientific scope of the project has been extended to cover also “Liquid-Solid phase transition triggered by dust cloud compression” (Topic T4) and “Stability and system dynamics (wave and energy) induced by active compression” (Topic T5) in complex plasma utilizing the PK-4 facility aboard the ISS as well as available COMPACT breadboard experiments. Concerning PK-4 experiments: We participated during the reporting period PK-4 ISS experiment campaigns in February 2022 (PK-4 campaign #14), January 2023 (PK-4 campaign #15), and February 2025 (PK-4 campaign #20). Our experiments aboard the International Space Station during all three campaigns were operated remotely from Auburn University (PI’s office) via two video feeds and additional voice communication channels to the ground control center in France (CADMOS) and our collaborators at Baylor University. The successful, complex communication arrangement was realized with the help of NASA, the European Space Agency (ESA), CADMOS, DLR and our local College of Science and Mathematics (COSAM) IT group. We are very grateful to all participants who made this wonderful way of conducting a space experiment (conveniently from a local university office) possible.

In the following, we give a short, topic-specific highlight summary of the activities/progress as archived in the reporting years (the activities funded under the research agreement RSA-JPL 1679198) towards its mission to gain new knowledge in the physics of complex (dusty) plasmas and to create a vision beyond the ongoing experiments towards future space experiments on complex plasmas in space (presumably aboard the ISS) with the experiment “COMPACT”, the Complex Plasma Facility. The research that has been conducted can be categorized in the initial three sub topics T1 to T3 and TX as specified in RSA-JPL-1571699 and the new activities T4, T5, TX as defined in the statement of work from the research agreement RSA-JPL-1679198:

Topic T1: Charging and interaction of micro-particles in a plasma. Topic T2: Ion-dust interaction and the thermal properties of dusty plasmas. Topic T3: Particle transport and heating in waves. Topic T4: Liquid-Solid phase transition triggered by dust cloud compression. Topic T5: Stability and system dynamics (wave and energy) induced by active compression. Topic TX: Future space experiments, administrative tasks, and developments.

Topic T1: To study the charging of micro-particles in a complex plasma, we followed two distinct approaches, leading in both cases to a successful conclusion of PhD projects with the submissions of the thesis during the reporting period. The first approach (to determine the average charge of the embedded micro-particles in a dusty/complex plasma) followed the overall charge neutrality assumption, that the electron, ion, and dust charge densities, when summed up, cancel each other. Within a PhD framework, an electron density measurement technique that is based on radio frequency (RF) attenuation/reflection measurement has been studied and developed – experimentally, theoretically, and by means of computer simulations. This technique has many advantages over Langmuir probe measurements (the most established electron density measurement) as it can be realized in a non-invasive mode, which is specifically of importance for a complex plasma, where a Langmuir probe leads to system disturbances. The radio frequency technique is not new; however, was not extensively studied or applied as a plasma diagnostic, probably as it requires the control and measurement of high frequency radio signals in the hundreds of MHz and low GHz range, which was difficult and costly decades ago, but is now addressable using off-the-shelf, dedicated, integrated electronic devices. Adding a non-invasive ion density measurement allows us to evaluate the charging state of embedded dust components in the complex plasma. Plans to integrate combined measurement techniques in the upcoming COMPACT experiment is a direct result of our work. The work was concluded with the PhD of Brandon Doyle in December 2023.

In a second approach to determine the particle charge in a plasma, we have studied the density variation of a streaming dust cloud in a dusty/complex plasma under the influence of a magnetic field. This work was mainly a theoretical and computer simulation endeavor and is expected to lead to a follow-up experimental realization in the future. The work predicts the existence of a complex plasma equivalent or similar effect as the classical “Hall-Effect” that can be utilized to determine the dust charge. Our theoretical and computational work on the complex plasma “Hall-Effect” was concluded with the PhD of Dylan Funk in 2023. Topics T2/T3 and T5: The study of topics T2,T3, and T5 has been led

**Task Progress:**

by the CoIs, Lori Scott, and some undergraduate students, in cooperation with Wittenberg University. During the reporting years, our work concentrated on the analysis of data that we obtained during campaign 7 of the PK-4 space experiment (executed July 2019; data received November 2019), data that was produced during a parabolic flight campaign under the umbrella of a cooperation with Giessen University in Germany, and data from PK-4 campaign 9 (executed February 2020). More recent activities have been utilizing data of our PK-4 participation during campaign 12 (executed June 2021), campaign 14 (executed Feb 2022), and campaign 15 (executed Jan 2023). The latter two were dedicated specifically to compression experiments of a complex plasma (Topic T4/T5). The most recent campaign from Feb 2025 is still under investigation. For all experiments, we studied how the energy distribution (the motion of the dust particles) and energy content stored in the structure of the system change shortly after the onset of an active disturbance to the system. Disturbances were realized by a variety of methods: a) by changing plasma production conditions (campaign 7) that actively change the dust-dust interactions and presumably, as the analysis suggests, immediately change the energy stored in the structure; b) by active manipulations using a laser beam (campaign 9) and practically shaking or pushing the dust component around; c) or by compressing the system actively via reducing the active plasma volume (campaign 12, 14, 20) similar to increasing the temperature in a gas by a fast (adiabatic) compression of the latter. While the analysis related to topic T5 is still ongoing, some work on T4/T5 has been recently finalized in the publication by L. Scott in the journal “Physics of Plasmas” as a follow-up to her PhD thesis.

Topic T4: The goal of this study is to understand how a rapid compression of a multi-particle system such as a complex plasma can trigger a phase transition and to answer questions such as: How much and how fast do we need to compress a system so that a liquid becomes a solid? How does the structure change under compression on the “quasi-atomic” scale? Do we always get a solid when we compress a system? To study these questions, we performed dedicated experiments with the PK-4 experiment aboard the ISS using a complex plasma as a model system. We compressed an obviously liquid-like complex plasma system by reducing the volume it can occupy while keeping the number of particles constant. The compressed system reacts by slowly changing its internal structure and at least partly transitions into a solid in most experiments. Supplemental experiments were performed with a next generation breadboard space experiment, a breadboard setup of the COMPACT facility. As the PK-4 experiment is subject to an ion flow induced heating that counteracts the crystallization process, the results of the latter are less clear in comparison with experiments performed in the COMPACT breadboard setup. A detailed analysis is in progress by a graduate student at Auburn University working full-time on this project since 2024. As part of his work, he has been developing computational techniques to efficiently characterize local structures such as liquid and solid (hexagonal, cubic) on the quasi-atomic scale from obtained image data received from the PK-4 or other similar experiments, such as the COMPACT breadboard setup. His specific goal is to study the compression induced evolution of the crystallization process (ratio of crystalline to liquid structure). The work on phase transitions such as solid-liquid, liquid-glass, and supercooled liquid-solid will expand on these types of activities in our future research endeavors.

Topic TX: In addition to the directly scientific-driven topics, our team also has been contributing to the definition of the scientific program, and the detailed design of a future, next generation complex plasma space experiment, the experiment facility “COMPACT”. Initial work was performed under a separate work agreement with NASA's Jet Propulsion Laboratory (JPL) on the US side and DLR on the German side, as part of the duties of a science definition team (SDT). The core responsibility of the SDT was to define the scientific mission of COMPACT based on the input of the overall, international complex plasma community, as well as of representatives with interests in solar system dust (lunar dust, Martian dust, or cometary dust). The collection of information at Auburn online-hosted workshops led to the science environment requirement document (SERD) for COMPACT. The work has continued under the umbrella of the here reported-on project for our members of the SDT. After formulation of the SERD, the SDT's work transitioned to defining the experiment design requirements that would allow the experiment facility to support the compiled scientific program. In close collaboration with industry and the German Aerospace Center, the SDT (that has been chaired in recent years by the PI of this project) has been supporting the project through its phase A and phase B (actual phase of project as of today) with a variety of activities that include design idea development and breadboard testing of critical components. One core interest of the Auburn group is to develop/promote COMPACT experiment capabilities in support of future studies based on astrophysical relevant material in COMPACT; in particular, performing transport and coagulation experiments using lunar surface equivalent material. As an example, Auburn is in the process of testing a potential UV light source for COMPACT that is necessary to simulate the solar light influence on the charging process of regolith (dust found on the Moon or asteroids).

Overall: We were able, during the reporting timeframe, to conclude major parts of our proposed, extended research plan as well as bring outstanding activities to a conclusion. The latter refers to the conclusion of the PhD submissions of Dylan Funk and Brandon Doyle, who were working mostly on the question T1, the charging of the dust. B. Doyle and U. Konopka have also been working on technical aspects for COMPACT, such as the radio frequency (RF) system and a potential RF plasma diagnostic system. These activities are still ongoing. The analysis of the most recent PK-4 experiment from Feb 2025, as well as some from former campaigns, are yet in process and expected to ultimately take a substantial part of the PhD work of M. Sake, an Auburn University (AU) graduate student. The work of M. Sake will continue beyond the here reported-on project utilizing in the future the COMPACT breadboard setup for additional experiments. COMPACT is specifically designed to support the specifically proposed compression experiments and thus is expected to give a better insight into the scientific question T4. Beyond the PhDs of B. Doyle and D. Funk in 2023, our work was disseminated during presentations at multiple conferences, such as the annual American Physical Society - Division of Plasma Physics (APS-DPP) meeting, the American Workshop of Dusty Plasmas (WDP-2024) in Minneapolis, MN, and the recent International Conference on the Physics of Dusty Plasmas (ICPDP) in May 2025 in Eindhoven, Netherlands. We also presented our ongoing activities at the NASA fundamental workshop series in 2023, 2024. Further, our work and related work was published in the form of several articles. The funding also allowed the PI/CoIs and graduate students to participate in a variety of outreach and service activities. For example, the PI co-funded, as faculty advisor, the Astronomical Society of Auburn University and has worked as advisor to the organization since then. He also has worked as the chair of the Science Definition Team for COMPACT and performed outreach at a local elementary school and other opportunities. Similar scenarios apply for the other CoIs. Concluding, we are very grateful to the agencies who have created for us opportunities to study fundamental scientific problems, and to participate in technological development and scientific exchange, by funding this project.

An outlook beyond PK-4: The PK-4 experiment is expected to soon approach its end of lifetime (EOL). Thus, we do not expect to perform additional experiments with PK-4. The data from PK-4, that we obtained during this project, will continue to flow into our future work; for example, as comparative experiments for the dust cloud compression experiments that we plan to conduct using the COMPACT breadboard setup, and in the near future also with the COMPACT space experiment. Our focus will be to support the development of COMPACT with breadboard activities

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| | and to utilize the new experiment for our research to explore collective phenomena from the atomic scale to the properties of the continuous media, especially addressing questions on structural transitions (phase transitions) and flow kinetics (turbulence) for which COMPACT is optimized. We expect also to continue our former activities to study astrophysical relevant dust environments using lunar equivalent dust or even grown ice particles to address cometary physics. Of particular interest is the charging in a plasma/UV light environment and the resulting transport of the particles across surfaces that might be a cause of erosion of crater structures on the lunar surface. The controlled transport of lunar/Martian dust is also a potential research topic that can lead to dust mitigation applications for space explorations to the Moon, Mars, or other solar system bodies. |
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